

23102A0007- Aarti Sakpal

Lab 3: Multi-Source Retail Sales Data Integration and Analysis Objective

To import, clean, integrate, analyze, and store heterogeneous retail sales data from CSV, JSON, and Excel sources using R and SQLite.

Dataset

UCI Online Retail Dataset

The original dataset contains transaction-level retail sales information.

Data Sources Created transactions.csv — InvoiceNo, StockCode, CustomerID, Quantity, InvoiceDate
products.json — StockCode, Description, UnitPrice customers.xlsx — CustomerID, Country

```
1 install.packages(c(
2   "tidyverse",
3   "jsonlite",
4   "readxl",
5   "writexl",
6   "DBI",
7   "RSQLite"
8 ))
```

Installing packages into '/usr/local/lib/R/site-library'
(as 'lib' is unspecified)

```
1 library(tidyverse)
2 library(jsonlite)
3 library(readxl)
4 library(writexl)
5 library(DBI)
6 library(RSQLite)
```

```
— Attaching core tidyverse packages — tidyverse 2.0.0 —
✓ dplyr      1.2.1    ✓ readr      2.2.0
✓ forcats    1.0.1    ✓ stringr    1.6.0
✓ ggplot2    4.0.3    ✓ tibble     3.3.1
✓ lubridate  1.9.5    ✓ tidyr      1.3.2
✓ purrr      1.2.2
```

```
— Conflicts — tidyverse_conflicts() —
```

```
✗ dplyr::filter() masks stats::filter()
```

```
✗ dplyr::lag() masks stats::lag()
```

```
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

Attaching package: 'jsonlite'

The following object is masked from 'package:purrr':

```
flatten
```

Task 1: Data Import and Cleaning

The UCI Online Retail dataset was imported into R using read_excel().

The dataset was inspected for:

Missing values Duplicate records Invalid or zero quantities Invalid or zero unit prices Cleaning Decisions
Records with missing CustomerID were removed because they could not be reliably associated with a customer. Records with missing Description were removed because product identification was incomplete. Duplicate records were removed using distinct(). Records with Quantity <= 0 were removed because they do not represent valid sales transactions. Records with UnitPrice <= 0 were removed because they cannot contribute to valid sales revenue. Revenue was calculated as:

Revenue = Quantity × UnitPrice

```
1 download.file(
2   "https://archive.ics.uci.edu/static/public/352/online+retail.zip",
3   "online_retail.zip"
4 )
5
6 unzip("online_retail.zip")
```

```
1 list.files()
```

```
'Online Retail.xlsx' · 'online_retail.zip' · 'sample_data'
```

```
1 retail_raw <- read_excel("Online Retail.xlsx")
```

```
1 head(retail_raw)
```

A tibble: 6 × 8							
InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country
<chr>	<chr>	<chr>	<dbl>	<dtm>	<dbl>	<dbl>	<chr>
536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	2010-12-01 08:26:00	2.55	17850	United Kingdom
536365	71053	WHITE METAL LANTERN	6	2010-12-01 08:26:00	3.39	17850	United Kingdom
536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	2010-12-01 08:26:00	2.75	17850	United Kingdom
536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	2010-12-01 08:26:00	3.39	17850	United Kingdom
536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	2010-12-01 08:26:00	3.39	17850	United Kingdom
SET 7							

```
1 dim(retail_raw)
```

541909 · 8

1 str(retail_raw)

```
tibble [541,909 × 8] (S3: tbl_df/tbl/data.frame)
 $ InvoiceNo  : chr [1:541909] "536365" "536365" "536365" "536365" ...
 $ StockCode  : chr [1:541909] "85123A" "71053" "84406B" "84029G" ...
 $ Description: chr [1:541909] "WHITE HANGING HEART T-LIGHT HOLDER" "WHITE METAL LANTERN" "CR
 $ Quantity   : num [1:541909] 6 6 8 6 6 2 6 6 6 32 ...
 $ InvoiceDate: POSIXct[1:541909], format: "2010-12-01 08:26:00" "2010-12-01 08:26:00" ...
 $ UnitPrice  : num [1:541909] 2.55 3.39 2.75 3.39 3.39 7.65 4.25 1.85 1.85 1.69 ...
 $ CustomerID : num [1:541909] 17850 17850 17850 17850 17850 ...
 $ Country    : chr [1:541909] "United Kingdom" "United Kingdom" "United Kingdom" "United Kin
```

1 summary(retail_raw)

InvoiceNo		StockCode		Description		Quantity	
Length	: 541909	Length	: 541909	Length	: 541909	Min.	: -80995.000
N.unique	: 25900	N.unique	: 4070	N.unique	: 4211	1st Qu.:	1.000
N.blank	: 0	N.blank	: 0	N.blank	: 0	Median	: 3.000
Min.nchar:	6	Min.nchar:	1	Min.nchar:	1	Mean	: 9.552
Max.nchar:	7	Max.nchar:	12	Max.nchar:	35	3rd Qu.:	10.000
				NAs	: 1454	Max.	: 80995.000

InvoiceDate		UnitPrice		CustomerID	
Min.	: 2010-12-01 08:26:00	Min.	: -11062.060	Min.	: 12346
1st Qu.:	2011-03-28 11:34:00	1st Qu.:	1.250	1st Qu.:	13953
Median	: 2011-07-19 17:17:00	Median	: 2.080	Median	: 15152
Mean	: 2011-07-04 13:34:57	Mean	: 4.611	Mean	: 15288
3rd Qu.:	2011-10-19 11:27:00	3rd Qu.:	4.130	3rd Qu.:	16791
Max.	: 2011-12-09 12:50:00	Max.	: 38970.000	Max.	: 18287
				NAs	: 135080

Country	
Length	: 541909
N.unique	: 38
N.blank	: 0
Min.nchar:	3
Max.nchar:	20

1 colSums(is.na(retail_raw))

InvoiceNo:	0	StockCode:	0	Description:	1454	Quantity:	0	InvoiceDate:	0	UnitPrice:	0
	0	CustomerID:	135080	Country:	0						

```
1 retail_clean <- retail_raw %>%
2   filter(
3     !is.na(Description),
4     !is.na(CustomerID)
5   )
```

1 dim(retail_clean)

406829 · 8

1 sum(duplicated(retail_clean))

5225

```
1 retail_clean <- retail_clean %>%  
2   distinct()
```

```
1 sum(duplicated(retail_clean))
```

0

```
1 dim(retail_clean)
```

401604 · 8

```
1 sum(retail_clean$Quantity <= 0)
```

8872

```
1 sum(retail_clean$Quantity == 0)
```

0

```
1 Start coding or generate with AI.
```

```
1 sum(retail_clean$Quantity < 0)
```

8872

```
1 retail_clean <- retail_clean %>%  
2   filter(Quantity > 0)
```

```
1 sum(retail_clean$Quantity <= 0)
```

0

```
1 dim(retail_clean)
```

392732 · 8

```
1 sum(retail_clean$UnitPrice <= 0)
```

40

```
1 sum(retail_clean$UnitPrice == 0)
```

40

```
1 retail_clean <- retail_clean %>%  
2   filter(UnitPrice > 0)
```

```
1 sum(retail_clean$UnitPrice <= 0)
```

0

1 dim(retail_clean)

392692 · 8

```
1 retail_clean <- retail_clean %>%
2   filter(UnitPrice > 0)
```

1 sum(retail_clean\$UnitPrice <= 0)

0

1 dim(retail_clean)

392692 · 8

Cleaning Results Stage Records Original dataset 541,909

After missing-value removal 406,829

After duplicate removal 401,604

After invalid quantity removal 392,732

After invalid price removal 392,692

Final Cleaning Verification

Missing Description = 0

Missing CustomerID = 0

Duplicate records = 0

Invalid Quantity = 0

Invalid UnitPrice = 0

Final records = 392,692

```
1 list(
2   Missing_Description = sum(is.na(retail_clean$Description)),
3   Missing_CustomerID = sum(is.na(retail_clean$CustomerID)),
4   Duplicate_Rows = sum(duplicated(retail_clean)),
5   Invalid_Quantity = sum(retail_clean$Quantity <= 0),
6   Invalid_UnitPrice = sum(retail_clean$UnitPrice <= 0),
7   Final_Rows = nrow(retail_clean)
8 )
```

```
$Missing_Description
0
$Missing_CustomerID
0
$Duplicate_Rows
0
$Invalid_Quantity
0
$Invalid_UnitPrice
0
$Final_Rows
392692
```

```
1 transactions <- retail_clean %>%
2   select(
3     InvoiceNo,
4     StockCode,
5     CustomerID,
6     Quantity,
7     InvoiceDate
8   )
```

```
1 head(transactions)
```

A tibble: 6 × 5

InvoiceNo	StockCode	CustomerID	Quantity	InvoiceDate
<chr>	<chr>	<dbl>	<dbl>	<dtm>
536365	85123A	17850	6	2010-12-01 08:26:00
536365	71053	17850	6	2010-12-01 08:26:00
536365	84406B	17850	8	2010-12-01 08:26:00
536365	84029G	17850	6	2010-12-01 08:26:00
536365	84029E	17850	6	2010-12-01 08:26:00
536365	22752	17850	2	2010-12-01 08:26:00

```
1 dim(transactions)
```

```
392692 · 5
```

```
1 write_csv(transactions, "transactions.csv")
```

```
1 list.files()
```

```
'Online Retail.xlsx' · 'online_retail.zip' · 'sample_data' · 'transactions.csv'
```

```
1 products <- retail_clean %>%
2   select(
3     StockCode,
4     Description,
5     UnitPrice
```

```
6 ) %>%
7 distinct(StockCode, .keep_all = TRUE)
```

```
1 head(products)
```

A tibble: 6 × 3

StockCode	Description	UnitPrice
<chr>	<chr>	<dbl>
85123A	WHITE HANGING HEART T-LIGHT HOLDER	2.55
71053	WHITE METAL LANTERN	3.39
84406B	CREAM CUPID HEARTS COAT HANGER	2.75
84029G	KNITTED UNION FLAG HOT WATER BOTTLE	3.39
84029E	RED WOOLLY HOTTIE WHITE HEART.	3.39
22752	SET 7 BABUSHKA NESTING BOXES	7.65

```
1 dim(products)
```

```
3665 × 3
```

```
1 write_json(
2   products,
3   "products.json",
4   pretty = TRUE,
5   auto_unbox = TRUE
6 )
```

```
1 list.files()
```

```
'Online Retail.xlsx' · 'online_retail.zip' · 'products.json' · 'sample_data' · 'transactions.csv'
```

Task 2: Data Integration

The three data sources were integrated using `left_join()`.

Join Keys Transactions + Products → StockCode Transactions + Customers → CustomerID Join Justification

`left_join()` was selected because the transaction dataset is the primary dataset. It ensures that all valid transaction records are retained while matching product and customer information is added where available.

Integration Results

Transactions 392,692 5

Products 3,665 3

Customers 4,338 2

Final integrated dataset 392,692 10

Unmatched Records

Unmatched products = 0

Unmatched customers = 0

The final integrated dataset contains:

InvoiceNo, StockCode, CustomerID, Quantity, InvoiceDate, Description, UnitPrice, Revenue, Country, Customer_Category

```
1 customers <- retail_clean %>%
2   select(
3     CustomerID,
4     Country
5   ) %>%
6   distinct(CustomerID, .keep_all = TRUE)
```

```
1 head(customers)
```

A tibble: 6 × 2

CustomerID	Country
<dbl>	<chr>
17850	United Kingdom
13047	United Kingdom
12583	France
13748	United Kingdom
15100	United Kingdom
15291	United Kingdom

```
1 dim(customers)
```

4338 × 2

```
1 write_xlsx(
2   customers,
3   "customers.xlsx"
4 )
```

```
1 list.files()
```

'customers.xlsx' · 'Online Retail.xlsx' · 'online_retail.zip' · 'products.json' · 'sample_data' · 'transactions.csv'

```
1 transactions_imported <- read_csv("transactions.csv")
```

Rows: 392692 Columns: 5

— Column specification —

Delimiter: ",",

chr (1): StockCode

dbl (3): InvoiceNo, CustomerID, Quantity

dtm (1): InvoiceDate

i Use ``spec()`` to retrieve the full column specification for this data.
i Specify the column types or set ``show_col_types = FALSE`` to quiet this message.

```
1 head(transactions_imported)
```

A tibble: 6 × 5

InvoiceNo	StockCode	CustomerID	Quantity	InvoiceDate
<dbl>	<chr>	<dbl>	<dbl>	<dtm>
536365	85123A	17850	6	2010-12-01 08:26:00
536365	71053	17850	6	2010-12-01 08:26:00
536365	84406B	17850	8	2010-12-01 08:26:00
536365	84029G	17850	6	2010-12-01 08:26:00
536365	84029E	17850	6	2010-12-01 08:26:00
536365	22752	17850	2	2010-12-01 08:26:00

```
1 products_imported <- fromJSON("products.json")
```

```
1 head(products_imported)
```

A data.frame: 6 × 3

	StockCode	Description	UnitPrice
	<chr>	<chr>	<dbl>
1	85123A	WHITE HANGING HEART T-LIGHT HOLDER	2.55
2	71053	WHITE METAL LANTERN	3.39
3	84406B	CREAM CUPID HEARTS COAT HANGER	2.75
4	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	3.39
5	84029E	RED WOOLLY HOTTIE WHITE HEART.	3.39
6	22752	SET 7 BABUSHKA NESTING BOXES	7.65

```
1 customers_imported <- read_excel("customers.xlsx")
```

```
1 head(customers_imported)
```

A tibble: 6 × 2

CustomerID	Country
<dbl>	<chr>
17850	United Kingdom
13047	United Kingdom
12583	France
13748	United Kingdom
15100	United Kingdom
15291	United Kingdom

```
1 cat("Transactions:", dim(transactions_imported), "\n")
2 cat("Products:", dim(products_imported), "\n")
3 cat("Customers:", dim(customers_imported), "\n")
```

Transactions: 392692 5
Products: 3665 3
Customers: 4338 2

```
1 sales_data <- transactions_imported %>%
2   left_join(
3     products_imported,
4     by = "StockCode"
5   )
```

```
1 dim(sales_data)
```

392692 · 7

1 head(sales_data)

A tibble: 6 × 7

InvoiceNo	StockCode	CustomerID	Quantity	InvoiceDate	Description	UnitPrice
<dbl>	<chr>	<dbl>	<dbl>	<dtm>	<chr>	<dbl>
536365	85123A	17850	6	2010-12-01 08:26:00	WHITE HANGING HEART T-LIGHT HOLDER	2.55
536365	71053	17850	6	2010-12-01 08:26:00	WHITE METAL LANTERN	3.39
536365	84406B	17850	8	2010-12-01 08:26:00	CREAM CUPID HEARTS COAT HANGER	2.75
536365	84029G	17850	6	2010-12-01 08:26:00	KNITTED UNION FLAG HOT WATER BOTTLE	3.39
536365	84029E	17850	6	2010-12-01 08:26:00	RED WOOLLY HOTTIE WHITE HEART.	3.39
536365	22752	17850	2	2010-12-01 08:26:00	SET 7 BABUSHKA NESTING BOXES	7.65

```
1 sum(is.na(sales_data$Description))
```

```
0
```

```
1 sum(is.na(sales_data$UnitPrice))
```

```
0
```

```
1 sales_data <- sales_data %>%
2   mutate(
3     Revenue = Quantity * UnitPrice
4   )
```

```
1 head(sales_data)
```

A tibble: 6 × 8

InvoiceNo	StockCode	CustomerID	Quantity	InvoiceDate	Description	UnitPrice	Revenue
<dbl>	<chr>	<dbl>	<dbl>	<dtm>	<chr>	<dbl>	<dbl>
536365	85123A	17850	6	2010-12-01 08:26:00	WHITE HANGING HEART T-LIGHT HOLDER	2.55	15.30
536365	71053	17850	6	2010-12-01 08:26:00	WHITE METAL LANTERN	3.39	20.34
536365	84406B	17850	8	2010-12-01 08:26:00	CREAM CUPID HEARTS COAT HANGER	2.75	22.00
536365	84029G	17850	6	2010-12-01 08:26:00	KNITTED UNION FLAG HOT WATER BOTTLE	3.39	20.34
536365	84029E	17850	6	2010-12-01 08:26:00	RED WOOLLY HOTTIE WHITE HEART.	3.39	20.34

```
1 sales_data %>%
2   select(Quantity, UnitPrice, Revenue) %>%
3   head()
```

A tibble: 6 × 3

Quantity	UnitPrice	Revenue
<dbl>	<dbl>	<dbl>
6	2.55	15.30
6	3.39	20.34
8	2.75	22.00
6	3.39	20.34
6	3.39	20.34
2	7.65	15.30

```
1 final_data <- sales_data %>%
2   left_join(
3     customers_imported,
4     by = "CustomerID"
5   )
```

```
1 dim(final_data)
```

392692 · 9

```
1 head(final_data)
```

A tibble: 6 × 9

InvoiceNo	StockCode	CustomerID	Quantity	InvoiceDate	Description	UnitPrice	Revenue	Country
<dbl>	<chr>	<dbl>	<dbl>	<dtm>	<chr>	<dbl>	<dbl>	<chr>
536365	85123A	17850	6	2010-12-01 08:26:00	WHITE HANGING HEART T-LIGHT HOLDER	2.55	15.30	United Kingdom
536365	71053	17850	6	2010-12-01 08:26:00	WHITE METAL LANTERN	3.39	20.34	United Kingdom
536365	84406B	17850	8	2010-12-01 08:26:00	CREAM CUPID HEARTS COAT HANGER	2.75	22.00	United Kingdom
536365	84029G	17850	6	2010-12-01 08:26:00	KNITTED UNION FLAG HOT WATER BOTTLE	3.39	20.34	United Kingdom
536365	84029E	17850	6	2010-12-01 08:26:00	RED WOOLLY HOTTIE WHITE HEART.	3.39	20.34	United Kingdom
536365	22752	17850	2	2010-12-01 08:26:00	SET 7 BABUSHKA NESTING	7.65	15.30	United Kingdom

```
1 sum(is.na(final_data$Country))
```

0

```
1 final_data %>%
2   filter(is.na(Country)) %>%
3   select(CustomerID) %>%
4   distinct() %>%
5   head()
```

A tibble: 0 × 1

CustomerID

<dbl>

Task 3: Sales and Customer Analysis

3.1 Total Sales Revenue

The total revenue generated from the cleaned dataset is approximately:

9,546,219

3.2 Top 5 Products by Revenue Rank Product Revenue

1 PAPER CRAFT, LITTLE BIRDIE 168,469.60

2 REGENCY CAKESTAND 3 TIER 135,495.30

3 WHITE HANGING HEART T-LIGHT HOLDER 93,745.65

4 MEDIUM CERAMIC TOP STORAGE JAR 81,032.64

5 JUMBO BAG RED RETROSPOT 76,028.70

Observation: PAPER CRAFT, LITTLE BIRDIE generated the highest revenue among the products.

```
1 total_revenue <- final_data %>%
2   summarise(
3     Total_Revenue = sum(Revenue, na.rm = TRUE)
4   )
5
6 total_revenue
```

A tibble: 1 × 1

Total_Revenue

<dbl>

9546219

```
1 top_products <- final_data %>%
2   group_by(StockCode, Description) %>%
3   summarise(
4     Total_Revenue = sum(Revenue, na.rm = TRUE),
5     .groups = "drop"
6   ) %>%
7   arrange(desc(Total_Revenue)) %>%
8   slice_head(n = 5)
```

```
1 top_products
```

A tibble: 5 × 3

StockCode	Description	Total_Revenue
<chr>	<chr>	<dbl>
23843	PAPER CRAFT , LITTLE BIRDIE	168469.60
22423	REGENCY CAKESTAND 3 TIER	135495.30
85123A	WHITE HANGING HEART T-LIGHT HOLDER	93745.65
23166	MEDIUM CERAMIC TOP STORAGE JAR	81032.64
85099B	JUMBO BAG RED RETROSPOT	76028.70

```

1 top_countries <- final_data %>%
2   group_by(Country) %>%
3   summarise(
4     Total_Revenue = sum(Revenue, na.rm = TRUE),
5     .groups = "drop"
6   ) %>%
7   arrange(desc(Total_Revenue)) %>%
8   slice_head(n = 5)

```

3.3 Top 5 Countries by Revenue Rank Country Revenue

1 United Kingdom 7,868,408.70

2 Netherlands 329,130.80

3 EIRE 287,260.90

4 Germany 233,804.10

5 France 203,628.30

Observation: The United Kingdom is the dominant market in terms of revenue.

```
1 top_countries
```

A tibble: 5 × 2

Country	Total_Revenue
<chr>	<dbl>
United Kingdom	7868408.7
Netherlands	329130.8
EIRE	287260.9
Germany	233804.1
France	203628.3

```

1 top_customers <- final_data %>%
2   group_by(CustomerID) %>%
3   summarise(
4     Total_Purchase_Value = sum(Revenue, na.rm = TRUE),
5     .groups = "drop"
6   ) %>%

```

```
7 arrange(desc(Total_Purchase_Value)) %>%
8 slice_head(n = 5)
```

3.4 Top 5 Customers by Purchase Value Rank CustomerID Purchase Value

1	18102	383,153.00
2	14646	323,767.50
3	17450	171,051.90
4	16446	168,472.50
5	14911	155,092.00

Observation: Customer 18102 generated the highest total purchase value.

```
1 top_customers
```

A tibble: 5 × 2

CustomerID	Total_Purchase_Value
<dbl>	<dbl>
18102	383153.0
14646	323767.5
17450	171051.9
16446	168472.5
14911	155092.0

```
1 customer_value <- final_data %>%
2   group_by(CustomerID) %>%
3   summarise(
4     Total_Purchase_Value = sum(Revenue, na.rm = TRUE),
5     .groups = "drop"
6   )
```

```
1 nrow(customer_value)
```

4338

```
1 quantile(
2   customer_value$Total_Purchase_Value,
3   probs = c(0.25, 0.50, 0.75),
4   na.rm = TRUE
5 )
```

25%: 317.835 50%: 703.57 75%: 1744.9075

```
1 q1 <- 317.835
2 q2 <- 703.57
3 q3 <- 1744.9075
4
5 customer_value <- customer_value %>%
6   mutate(
```

```

7   Customer_Category = case_when(
8     Total_Purchase_Value <= q1 ~ "Low Value",
9     Total_Purchase_Value <= q2 ~ "Medium Value",
10    Total_Purchase_Value <= q3 ~ "High Value",
11    TRUE ~ "Premium"
12  )
13 )

```

```

1 customer_value %>%
2   count(Customer_Category)

```

A tibble: 4 × 2

Customer_Category	n
<chr>	<int>
High Value	1084
Low Value	1085
Medium Value	1084
Premium	1085

Customer Value Classification

Customers were classified using quartile-based thresholds calculated from total purchase values.

Category Threshold

Low Value ≤ 317.835

Medium Value $317.835 - 703.57$

High Value $703.57 - 1,744.9075$

Premium $> 1,744.9075$

The classification was implemented using the `case_when()` function.

Customer Distribution

Category Customers

Low Value 1,085

Medium Value 1,084

High Value 1,084

Premium 1,085

Total 4,338

```

1 final_data <- final_data %>%
2   left_join(
3     customer_value %>%
4     select(
5       CustomerID,
6       Customer_Category
7     ),

```



```
8   by = "CustomerID"
9   )
```

1 head(final_data)

A tibble: 6 × 10

InvoiceNo	StockCode	CustomerID	Quantity	InvoiceDate	Description	UnitPrice	Revenue	Country
<dbl>	<chr>	<dbl>	<dbl>	<dtm>	<chr>	<dbl>	<dbl>	<chr>
536365	85123A	17850	6	2010-12-01 08:26:00	WHITE HANGING HEART T-LIGHT HOLDER	2.55	15.30	United Kingdom
536365	71053	17850	6	2010-12-01 08:26:00	WHITE METAL LANTERN	3.39	20.34	United Kingdom
536365	84406B	17850	8	2010-12-01 08:26:00	CREAM CUPID HEARTS COAT HANGER	2.75	22.00	United Kingdom
536365	84029G	17850	6	2010-12-01 08:26:00	KNITTED UNION FLAG HOT WATER BOTTLE	3.39	20.34	United Kingdom
536365	84029E	17850	6	2010-12-01 08:26:00	RED WOOLLY HOTTIE WHITE HEART.	3.39	20.34	United Kingdom
536365	22752	17850	2	2010-12-01 08:26:00	SET 7 BABUSHKA NESTING BOXES	7.65	15.30	United Kingdom

```
1 dim(final_data)
```

392692 · 10

```
1 market_performance <- final_data %>%
2   group_by(Country) %>%
3   summarise(
4     Total_Revenue = sum(Revenue, na.rm = TRUE),
5     .groups = "drop"
6   ) %>%
7   arrange(desc(Total_Revenue))
```

```
1 head(market_performance, 5)
```

A tibble: 5 × 2

Country	Total_Revenue
<chr>	<dbl>
United Kingdom	7868408.7
Netherlands	329130.8
EIRE	287260.9
Germany	233804.1
France	203628.3

Market Performance High-Performing Market

United Kingdom

Revenue: 7,868,408.70

The United Kingdom is the strongest market and contributes substantially more revenue than the other countries.

Underperforming Market

Saudi Arabia

Revenue: 145.92

Saudi Arabia generated the lowest revenue among the countries in the dataset. The company could investigate demand, marketing reach, product availability, and distribution opportunities in this market.

1 tail(market_performance, 5)

A tibble: 5 × 2

Country	Total_Revenue
<chr>	<dbl>
Brazil	1184.43
RSA	971.82
Czech Republic	875.46
Bahrain	544.80
Saudi Arabia	145.92

Task 4: SQLite Database

The cleaned and integrated dataset was stored in a SQLite database:

Database: retail_sales.db

Table: retail_sales

The table contains:

392,692 records

10 columns

Database Columns

InvoiceNo

StockCode

CustomerID

Quantity

InvoiceDate

Description

UnitPrice

Revenue

Country

Customer_Category

SQLite Verification

The final dataset was successfully stored in the SQLite database retail_sales.db under the table retail_sales.

Database file exists: TRUE

Table name: retail_sales

Total records: 392,692

Total columns: 10

SQLite connection closed successfully after querying.

```
1 con <- dbConnect(  
2   SQLite(),  
3   "retail_sales.db"  
4 )
```

```
1 dbIsValid(con)
```

TRUE

```
1 dbWriteTable(  
2   con,  
3   "retail_sales",  
4   final_data,  
5   overwrite = TRUE  
6 )
```

```
1 dbListTables(con)
```

```
'retail_sales'
```

```
1 dbListFields(con, "retail_sales")
```

```
'InvoiceNo' · 'StockCode' · 'CustomerID' · 'Quantity' · 'InvoiceDate' · 'Description' · 'UnitPrice' · 'Revenue' ·  
'Country' · 'Customer_Category'
```

```
1 dbGetQuery(  
2   con,  
3   "SELECT COUNT(*) AS Total_Rows FROM retail_sales"  
4 )
```

A data.frame:

1 × 1

Total_Rows

<int>

392692

```
1 query1 <- "  
2 SELECT  
3   CustomerID,  
4   SUM(Revenue) AS Total_Revenue  
5 FROM retail_sales  
6 GROUP BY CustomerID  
7 ORDER BY Total_Revenue DESC  
8 LIMIT 5  
9 "
```

SQL Query 1: Top 5 Customers by Revenue

The SQL query was executed from R using DBI and RSQLite.

```
1 top5_customers_sql <- dbGetQuery(con, query1)  
2  
3 top5_customers_sql
```

A data.frame: 5 × 2

CustomerID **Total_Revenue**

<dbl>

<dbl>

18102 383153.0

14646 323767.5

17450 171051.9

16446 168472.5

14911 155092.0

SQL Query 2: Revenue by Country

```
1 query2 <- "  
2 SELECT
```

```
3 Country,
4 SUM(Revenue) AS Total_Revenue
5 FROM retail_sales
6 GROUP BY Country
7 ORDER BY Total_Revenue DESC
8 "
9
10 revenue_by_country_sql <- dbGetQuery(con, query2)
11
12 head(revenue_by_country_sql, 10)
```

A data.frame: 10 × 2

	Country	Total_Revenue
	<chr>	<dbl>
1	United Kingdom	7868408.65
2	Netherlands	329130.77
3	EIRE	287260.87
4	Germany	233804.06
5	France	203628.28
6	Australia	157316.33
7	Spain	62080.85
8	Switzerland	57268.14
9	Belgium	43098.99
10	Japan	42082.88

```
1 dbDisconnect(con)
```

```
1 dbIsValid(con)
```

FALSE

```
1 file.exists("retail_sales.db")
```

TRUE

```
1 file.info("retail_sales.db")$size
```

37822464

Three Important Business Insights

- 1. United Kingdom is the dominant market

The United Kingdom generated approximately 7.87 million in revenue, making it the strongest market by a significant margin.